

Chapter SIX

Computer System Security

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- Basics of Data privacy
- An overview of computer system security
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What is Data Privacy?

- Data privacy generally means the ability of a person to determine for themselves :
 - ❑ When
 - ❑ How, and
 - ❑ To what extent personal information about them is shared with or communicated to others.
- This personal information can be one's name, location, contact information, or online or real-world behavior.
- Just as someone may wish to exclude people from a private conversation, many online users want to control or prevent certain types of personal data collection.

Why is Data Privacy Important?

- Data privacy is also important because in order for individuals to be willing to **engage online**,
- They have to trust that their personal data will be **handled with care**.
- Organizations use **data protection practices** to demonstrate to their **customers** and **users** that they can be trusted with their personal data.
- Personal data can be **misused** in a number of ways if it is not kept private or if people don't have the ability to control **how their information** is used:

Why is Data Privacy Important? (Cont.)

- Criminals can use personal data to defraud or harass users
- Entities may sell personal data to advertisers or other outside parties without user consent, which can result in users receiving unwanted marketing or advertising
- When a person's activities are tracked and monitored, this may restrict their
 - ability to express themselves freely, especially under repressive governments

What are Fair Information Practices?

- Many of the existing data protection laws are based on foundational privacy principles and practices, such as those laid out in the Fair Information Practices.
- The Fair Information Practices is a set of guidelines for data collection and usage. Some of the practices are
- **Collection limitation**: There should be limits to how much personal data can be collected
- **Data quality**: Personal data, when collected, should be accurate and related to the purpose it is being used for
- **Purpose specification**: The use for personal data should be specified

What are Fair Information Practices? (Cont.)

- Use limitation: Data should not be used for purposes other than what was specified
- Security safeguards: Data should be kept secure
- Openness: Personal data collection and usage should not be kept secret from individuals
- Individual participation: Individuals have a number of rights, including the right to know who has their personal data, to have their data communicated to them, to know why a request for their data is denied, and to have their personal data corrected or erased
- Accountability: Anyone who collects data should be held accountable for implementing these principles

Computer Security

- Computer security basically is the protection of computer systems and information from harm, theft, and unauthorized use.
- It is the process of preventing and detecting unauthorized use of your computer system.
- There are various types of computer security which is widely used to protect the valuable information of an organization
- One way to ascertain the similarities and differences among Computer Security is by asking what is being secured. For example,
- Information security is securing information from unauthorized access, modification & deletion
- Application Security is securing an application by building security features to prevent from Cyber Threats such as SQL injection, DoS attacks, data breaches and etc.

Computer Security (Cont.)

- **Computer Security** means securing a standalone machine by keeping it updated and patched
- **Network Security** is by securing both the software and hardware technologies
- **Cybersecurity** is defined as protecting computer systems, which communicate over the computer networks
- It's important to understand the distinction between these words, though there isn't necessarily a clear consensus on the meanings and the degree to which they overlap or are interchangeable.

Components of Computer System

- Computer security is mainly concerned with three main areas also known as **the CIA Triad**



- **Confidentiality** is ensuring that information is **available only** to the **intended audience** (**prevent** sensitive information from unauthorized access attempts.)
- **Integrity** involve maintaining consistency, accuracy and trustworthiness of data over its entire lifecycle (data cannot be altered by unauthorized person)
- **Availability** information should be **consistently** and **readily** accessible for authorized parties.

Computer Security Threats

- Computer security threats are possible dangers that can possibly hamper the normal functioning of your computer.
- In the present age, cyber threats are constantly increasing as the world is going digital.
- The most harmful types of computer security threats are:

Virus

- Is a malicious program which is loaded into the user's computer without user's knowledge.
- It replicates itself and infects the files and programs on the user's PC.
- The ultimate goal of a virus is to ensure that the victim's computer will never be able to operate properly or even at all.

Computer Security Threats (Cont.)

Computer Worm

- A computer worm is a software program that can **copy itself from one computer to another**, without **human interaction**.
- The potential risk here is that it will use up your computer **hard disk space** because a worm can replicate in **greater volume** and **with great speed**.

Phishing

- Disguising as a trustworthy person or business, phishers attempt to steal sensitive financial or personal information through **fraudulent email** or **instant messages**.
- Phishing is unfortunately very easy to **execute**.
- You are deluded into thinking it's the **legitimate mail** and you may enter **your personal information**.

Computer Security Threats

Rootkit

- A rootkit is a computer program designed to provide continued privileged access to a computer while actively hiding its presence.
- Once a rootkit has been installed, the controller of the rootkit will be able to remotely execute files and change system configurations on the host machine.

Keylogger

- Also known as a keystroke logger
- Keyloggers can track the real-time activity of a user on his computer.
- It keeps a record of all the keystrokes made by user keyboard.
- Keylogger is also a very powerful threat to steal people's login credential such as username and password.

Computer Security Policies

- In a way they are the regulatory of the behaviors of your employees towards the use of technology in the workplace,
- That can minimize the risk of being hacked, information leak, internet bad usage and it also ensures safeguarding of company resources.
- In real life you will notice the employees of your organization will always tend to click on bad or virus infected URL's or email attachments with viruses.
- Role of the Security Policy in Setting up Protocols

Computer Security Policies (Cont.)

Following are some pointers which help in setting you protocols for the security policy of an organization.

- Who should have access to the system?
- How it should be configured?
- How to communicate with third parties or systems?
- Policies are divided in two categories –
 - i. User policies
 - ii. IT policies

Computer Security Policies (Cont.)

- User policies generally define the **limit** of the users towards the computer resources in a workplace.
- For example, what are they allowed to **install in their computer**, if they can use **removable storages**.
- Whereas, IT policies are designed for IT department, to **secure the procedures** and **functions of IT fields**.
- Some of the **IT policies**

General Policies – This is the policy which defines the rights of the staff and access level to the systems.

- Generally, it is included even in the communication protocol as a preventive measure in case there are any disasters.

Server Policies – This defines **who should have access to the specific server** and with what rights.

- Which software's should be **installed**, **level of access to internet**, **how they should be updated**.

Computer Security Policies (Cont.)

Firewall Access and Configuration Policies – It defines who should have access to the firewall and what type of access, like monitoring, rules change.

- Which **ports** and **services** should be allowed and if it should be inbound or outbound.
- **Backup Policies** – It defines
 - ☐ Who is the **responsible** person for backup,
 - ☐ What should be the **backup**,
 - ☐ Where it should be **backed up**,
 - ☐ How long it should be **kept** and the **frequency of the backup**.

VPN Policies – These policies generally go with the **firewall policy**

- It defines those users who should have a **VPN access** and **with what rights**.
- For site-to-site connections with partners, it defines the access level of the partner to your network, type of encryption to be set

Structure of a Security Policy

- When you compile a **security policy** you should have in mind a basic structure in order to make something practical.
- Some of the **main points** which have to be taken into consideration are –
 - ☐ **Description** of the **Policy** and **what is the usage for?**
 - ☐ Where this **policy** should be applied?
 - ☐ **Functions** and **responsibilities** of the employees that are affected by this policy.
 - ☐ Procedures that are **involved** in this **policy**.
 - ☐ **Consequences** if the policy is **not compatible** with company standards.

Types of Policies

- **Permissive Policy** – It is a **medium restriction policy** where we as an administrator **block** just some well-known ports of malware regarding **internet access** and just some exploits are taken in consideration.
- **Prudent Policy** – This is a **high restriction policy** where everything is blocked regarding the internet access, just a small list of websites are allowed, and now extra services are allowed in computers to be installed and logs are maintained for every user.
- **Acceptance User Policy** – This policy regulates the **behavior of the users towards a system** or **network** or even a **webpage**, so it is explicitly said what a user **can do** and **cannot in a system**. Like are they allowed to share access codes, can they share resources, etc.

Types of Policies (Cont.)

- **Information Protection Policy** – This policy is to regulate **access to information, how to process information**, how to **store** and how it should be **transferred**.
- **Remote Access Policy** – This policy is mainly for big companies where the user and their branches are outside their headquarters.
- It tells what should the users access, when they can work and on which software/ protocol like SSH, VPN.

Computer Security Practices

- Computer security threats are becoming **relentlessly inventive these days**.
- There is much need for one to arm **oneself with information** and **resources** to safeguard against these complex and growing computer security threats and stay safe online.

Some **preventive** steps you can take include:

- Installing **reliable, reputable** security and **anti-virus software**
- **Activating your firewall**, because a firewall acts as a **security** guard between the internet and your local area network
- Stay **up-to-date on the latest software** and news surrounding your devices and perform **software updates** as soon as they become available
- Avoid clicking on **email attachments unless you know the source**
- Change **passwords regularly**, using a unique combination of **numbers, letters** and **case types**

Computer Security Practices (Cont.)

- Use the internet with **caution** and **ignore pop-ups**, drive-by downloads while surfing
- Taking the time to research the basic **aspects of computer security** and **educate yourself** on evolving **cyber-threats**
- Perform **daily full system scans** and create a **periodic system backup** schedule to ensure your data is **retrievable** should something happen to your computer.